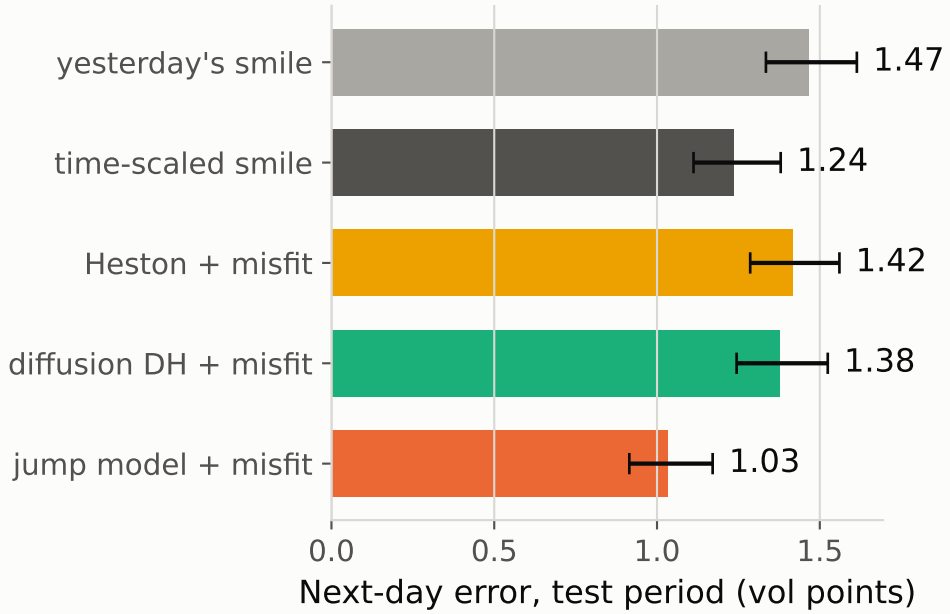
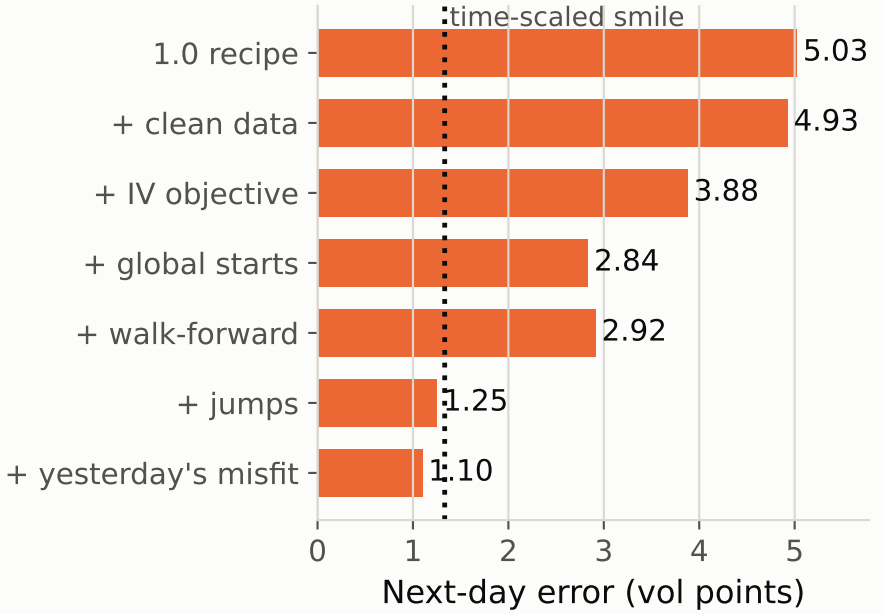
Double Heston 2.0 on NIFTY 50 options: summary of the results

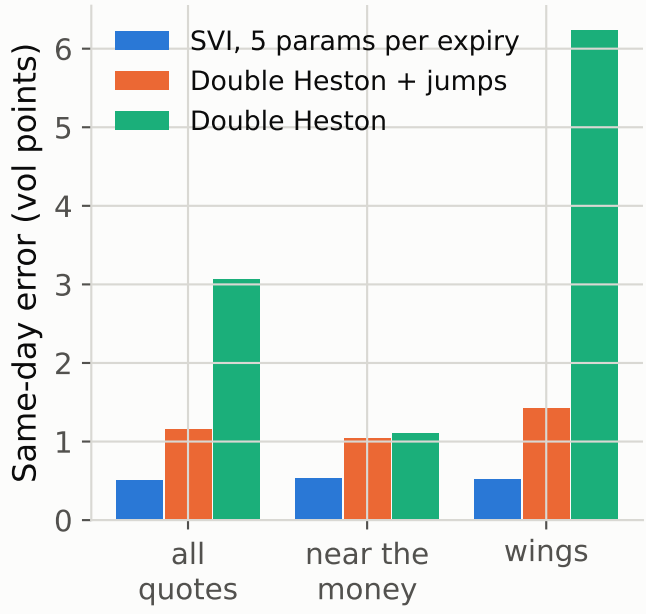
Forecast error



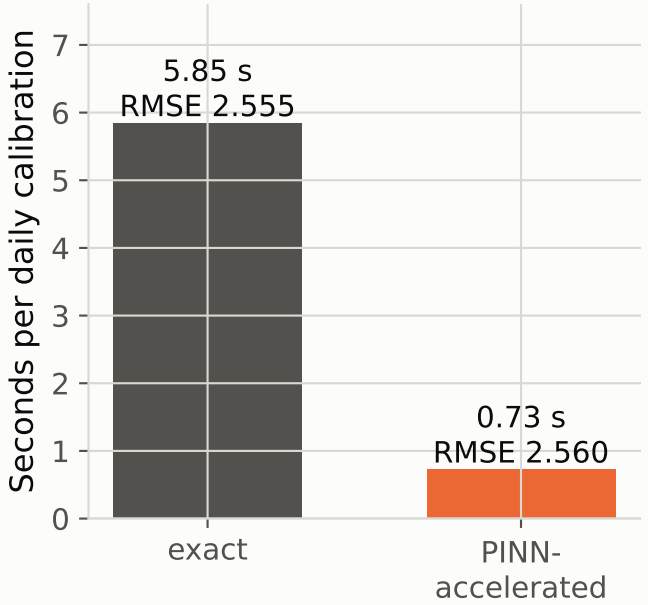
From the 1.0 recipe to 2.0



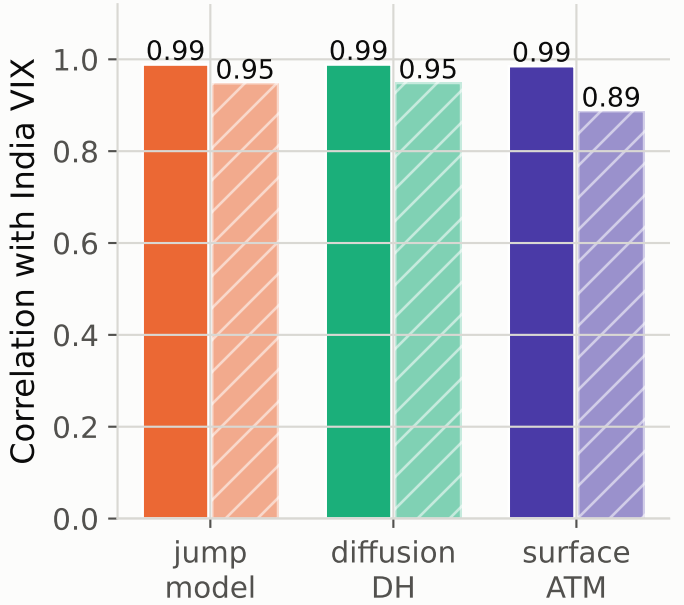
Fit quality against SVI



Faster calibration, same fit (8.0x)



Link to India VIX



solid: levels, 2016-2026; hatched: daily changes, test period

Key numbers

Jump + misfit vs best benchmark, test
1.034 vs 1.236 vol pts

Paired difference, 95% interval
-0.202 [-0.227, -0.181]

Project pricer vs COS, real chains
1.5e-06 vol pts at most

PINN surrogate vs exact pricer, mean error
0.049 vol pts

Surrogate speed-up, 128 parameter sets
21x on CPU, 79x on GPU

Held-out quotes: jump model vs diffusion
0.71 vs 2.49 vol pts